ADVANCED NLP · MONSOON 2026

Which parts of BM25 help learned top-k selection?

Tests in sparse retrieval, KV-cache page selection and expert routing

Team ANLP Doomsday

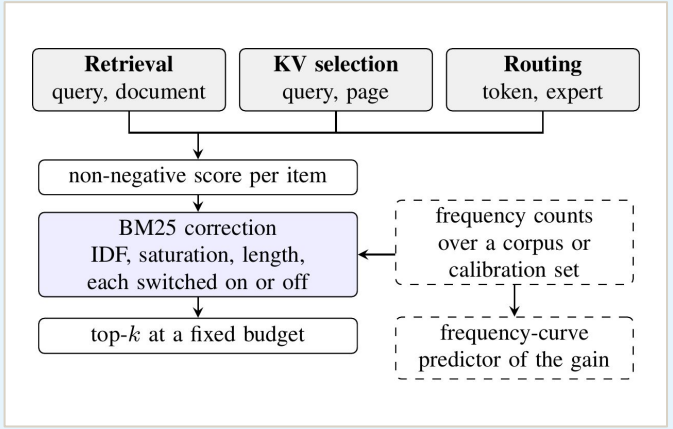


Figure 1 of our report: the three settings and where the BM25 correction sits

Three systems, one computation



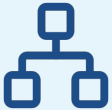
Retrieval

A query scores documents.
Keep the top-k documents.



KV-cache selection

A query scores KV pages (Quest).
Attend only to the top-k pages.



MoE routing

A token scores experts.
Send it to the top 8 of 64.

Only retrieval corrects the raw score, with BM25. Does that correction travel, and can we tell when it helps?

BM25 is three small fixes to a raw score



IDF

Down-weight features that show up everywhere in the collection.



Saturation

Stop rewarding repeated evidence past a point.



Length

Neither favour nor punish long documents.

$$S(q, d) = \Sigma_j \, x_{qj} \cdot w_j(\mathrm{IDF}) \cdot T(x_{dj}, \ell_d)$$

Each fix is on or off, so 8 configurations. **000** = raw inner product, **111** = BM25.

Built for words, not for learned features.

Prior work disagrees about whether it works

+0.20

nDCG@10 on sparse-autoencoder features of a dense retriever

−0.11

nDCG@10 on SPLADE-v3

The popular explanation: BM25 helps when feature frequencies are Zipf-like, as word frequencies are.

If that is true, the frequency curve should predict the gain before running anything. We tested exactly that.

Clavié et al. 2026 · Han et al. 2026

Four questions, four short answers

01	Does BM25 beat the raw inner product on learned sparse features?	**Yes: 162 of 182 rows mean nDCG@10 .036 → .158**
02	Which of IDF, saturation and length explains the gain?	**Mostly saturation (62%) which needs no collection statistics**
03	Can we predict the gain before measuring it?	**Saturation's, yes. The Zipf curve's, no.**
04	Does it carry over to KV-cache selection and MoE routing?	**No: IDF hurts both, and attention mass shows why**

STUDY DESIGN

One encoder, 14 sparse codes, 8 configurations



Text
queries and documents

Frozen E5
one dense encoder

Sparsify
7 bases, pooled or token

Score
8 BM25 configurations

nDCG@10
ranking quality

14
sparse representations

13
collections, English + Hindi, Bengali, Telugu

8
on/off configurations

1,358
rows, each with 8 nDCG@10 scores

Exact Shapley values split each gain. Paired bootstrap with Holm correction. Predictors validated on held-out families and corpora.

RIGOUR

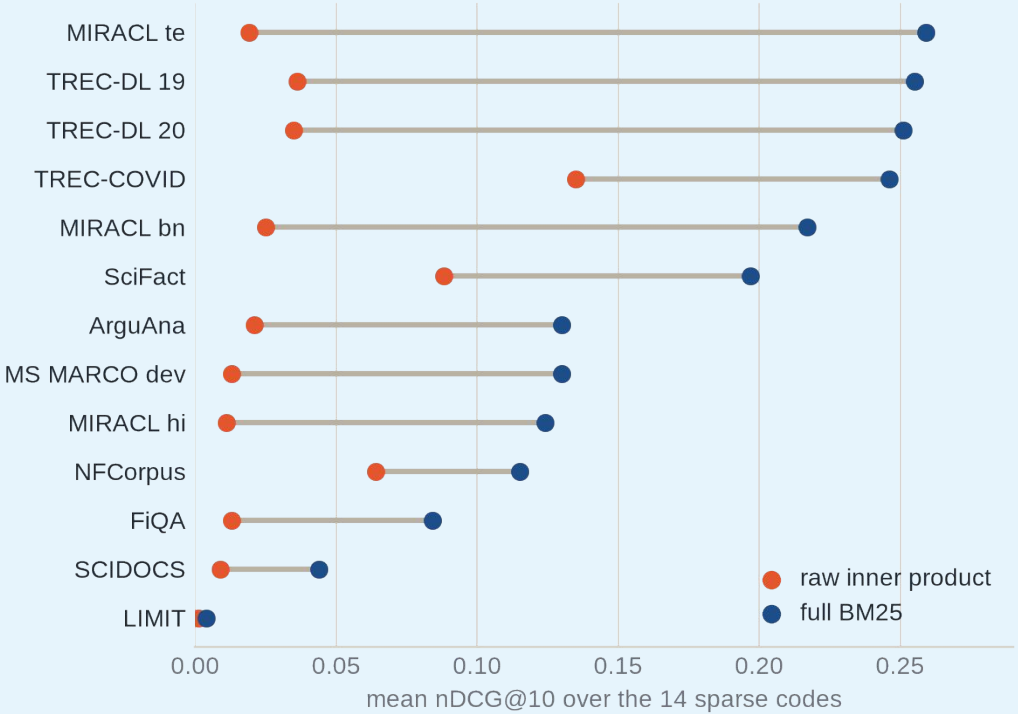
Tests were fixed before their results were read



Six registered tests, and each was fixed before its results were read.

So a prediction that could fail did fail, and we can report it.

BM25 lifts the sparse codes more than fourfold



89%

of rows improve when BM25 replaces the raw inner product.

The best code (token-level SAE) beats text BM25 on 6 of 13 collections.

Caveat: the raw score is below 0.01 in 81 rows, so nearly any fix helps.

RQ2 · ATTRIBUTION

Most of the gain needs no collection statistics

Share of the raw → BM25 gain, exact Shapley values over 182 primary rows



Uses only the document's own values. No collection, no labels.

Second largest. Helps the E5 codes, but hurts SPLADE-v3.

The part the Zipf story is about contributes least, and shrinks as supports grow.

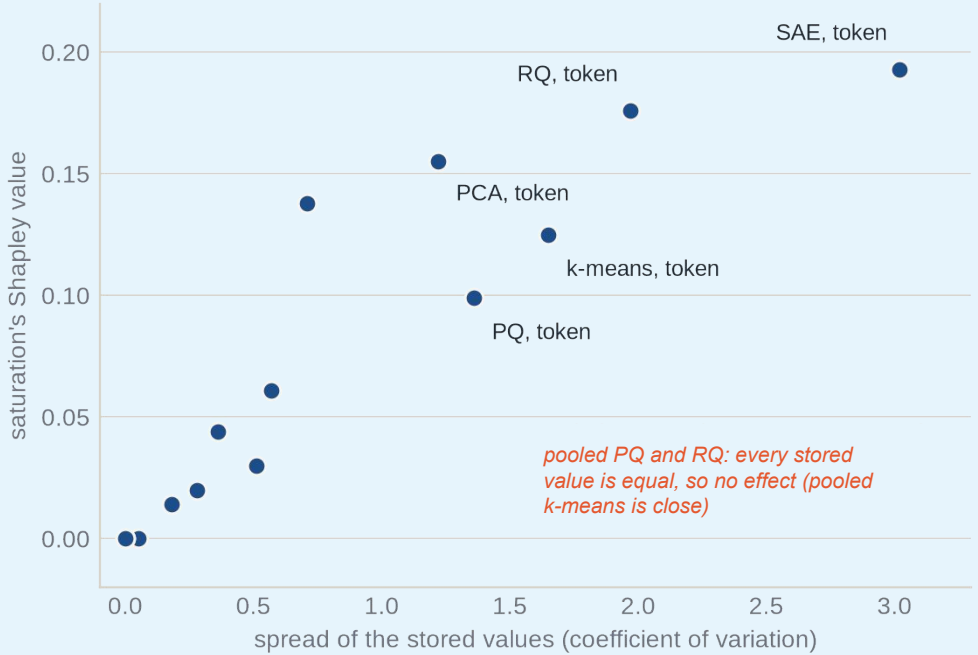
Collection statistics explain only 38% of the gain (95% CI 36 to 41%).

Saturation helps when stored values are uneven

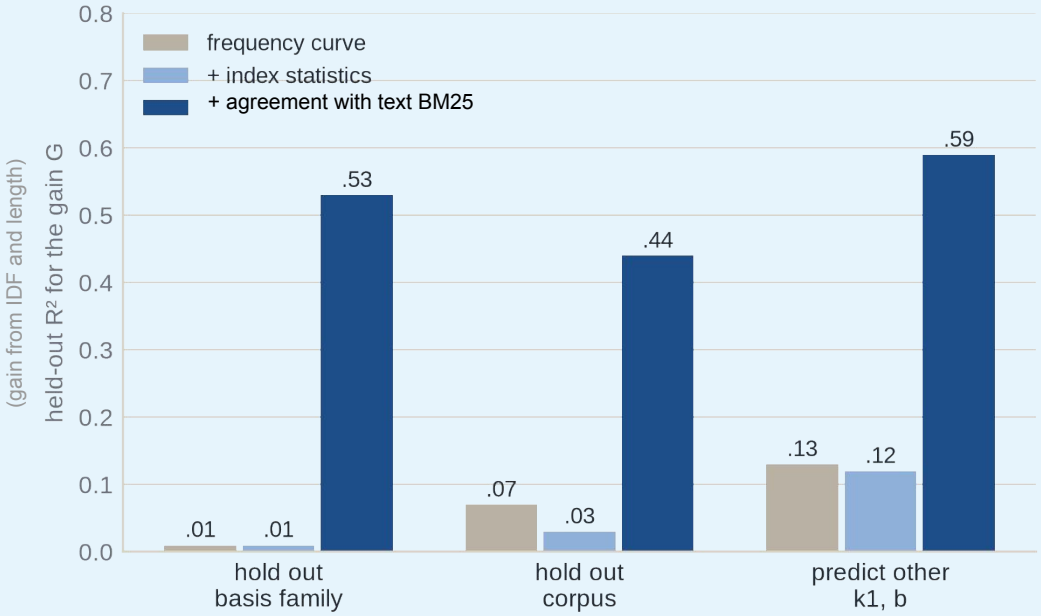
If every stored value were equal, saturation would rescale all scores by one factor and the ranking could not move.

More spread means more reordering. The prediction held on settings we had not seen (Spearman .87, .90) and across all 182 rows (.78). All six hypotheses pass.

Saturation alone lifts mean nDCG@10 from .036 to .137.



The Zipf curve doesn't predict the gain. Agreement with text BM25 does better.



.02–.04

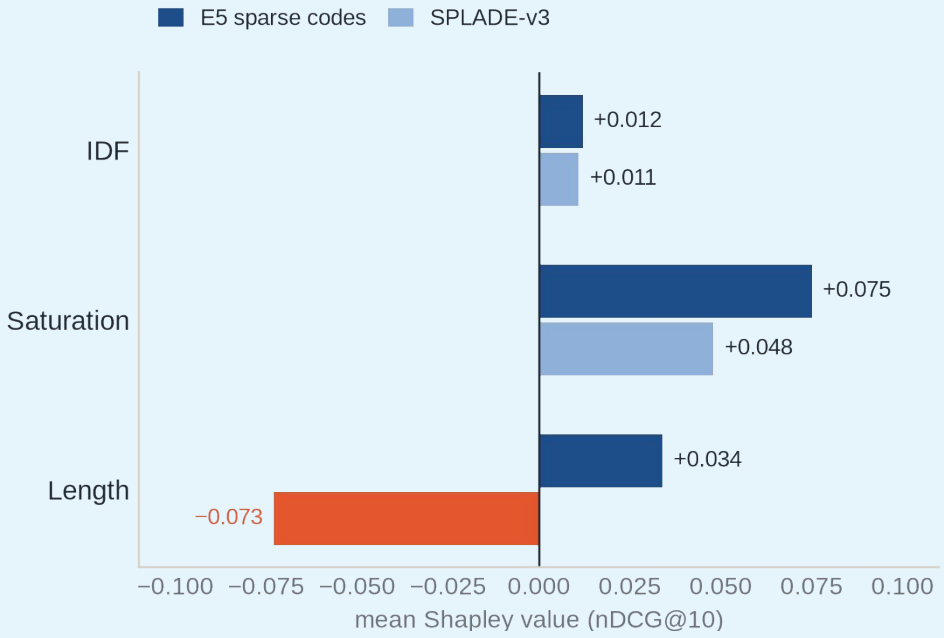
R^2 of the Zipf-curve predictor on held-out codes and corpora. Its sign is right 90% of the time, the same as always guessing 'BM25 helps'.

.44–.59

R^2 once agreement with text BM25 is added (dark bars).

Fitted models.
As a fixed rule, BM25 agreement is no better than a constant on new rows.

On SPLADE-v3, length normalisation is the problem



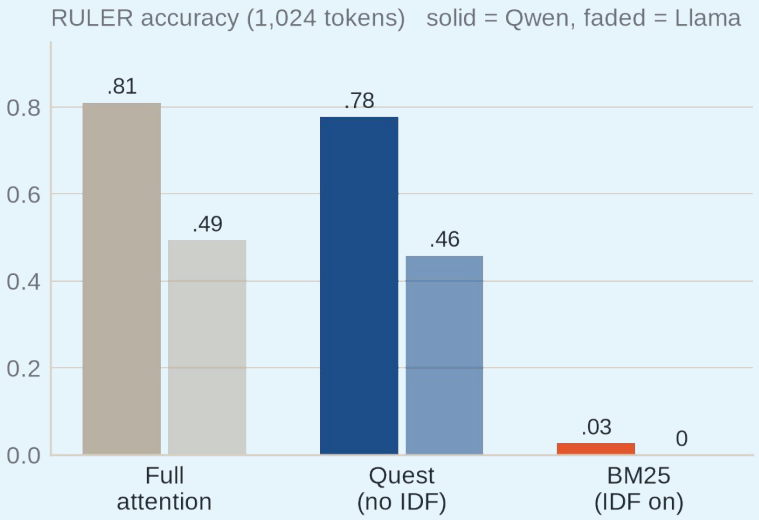
.523 → .509

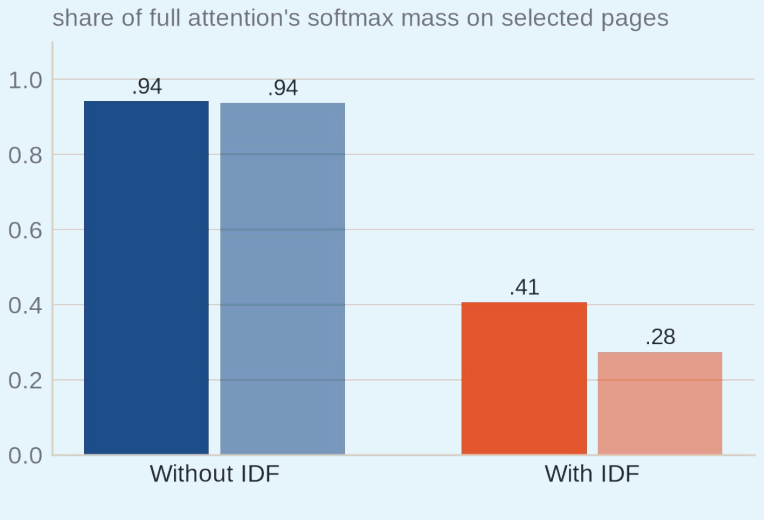
SPLADE-v3 nDCG@10 from the raw score to BM25, worse on 8 of 10 collections. Without length normalisation, BM25 slightly helps (.523 → .528).

Without labels, agreement with text BM25 picks the wrong setting here (right on 5 of 10). Agreement with dense E5 is right on 7 of 10.

The answer depends on the representation, not just the method.

In KV selection, IDF discards the pages attention needs





IDF down-weights the pages Quest selects often. Those are the pages the model needs. Significantly worse on all 24 Qwen rows and 18 of 23 Llama rows, better on none. Without labels, attention mass flags this on every task.

Honest caveat: partly by construction, since we define a page's frequency as its own selection rate.

MoE routing agrees, and why saturation can't help

PROPOSITION 1

With one score per item and IDF off, saturation selects the same top-k as no saturation.

Saturation is a strictly increasing function of f/ℓ, so it cannot reorder single-score items. Only IDF can, and here it reorders them wrongly.

✓ Confirmed: 001 to 011 match 000 on every example

OLMoE-1B-7B: MMLU accuracy over 57 subjects



WikiText-2 perplexity 7.49 → 7.88.
Worse on 52 of 57 subjects, better on 3.

SCORECARD

Six registered tests: 1 passed, 3 failed, 2 inconclusive

✓	**Saturation mechanism (H1 to H6)**	k1 and b sweep	All six pass: value spread predicts saturation's effect
⚖	**Frequency-curve predictor**	retrieval, 179 rows	R^2 .04 and .02, sign no better than a constant
✗	**Label-free rule**	SPLADE-v3, 10 collections	Wrong half the time with text BM25. Dense E5 gets 7 of 10
✗	**Frequency-curve predictor**	KV pages, Qwen and Llama	Wrong sign on every row (MAE 4.18, 4.00; limit 1.81)
⚖	**Label-free rule**	KV pages, Qwen and Llama	Picks the better setting on every row but a constant does too
✗	**Frequency-curve predictor**	routing, 57 subjects	Wrong sign on 52 of 55 subjects (MAE 3.98)

The saturation mechanism held; the Zipf-curve predictor predicted nothing in any of the three settings.

TAKEAWAYS

What this means beyond our grid

1 **Start with saturation**

Cheap, label-free, and alone it lifts nDCG@10 from .036 to .137.

2 **Don't port IDF blindly**

What counts as "frequency" decides it. In KV and routing, IDF cost accuracy (.78 to .03, .48 to .38).

3 **Zipf isn't enough**

Value spread tracks saturation. The frequency curve predicts almost nothing.

4 **Check without labels**

Attention mass flagged IDF's harm on every KV task.

Limits one encoder (E5), one 4K context with 100 examples per task, one MoE model.
Next frequencies counted over unrelated prompts.

Thank you

Questions?

4x

nDCG@10 from BM25 on sparse codes

62%

of it from saturation

.94 → .41

attention mass once IDF is on

github.com/avi1o1/ANLP-Doomsday

Aviral Gupta · Mohit Kumar Singh · Anirudh Sankar · Arihant Tripathy